

MID-STATE TECHNICAL COLLEGE • IT PROGRAMS

Creating and Managing NAT Policy Rules

Unit 7 • Information Security 2 (10-151-111)



Learning Outcomes

By the end of this unit, you will be able to:



Explain NAT in the flow logic

Describe when NAT policy is evaluated and when the translation is applied.



Configure source NAT

Translate private source addresses with static IP, dynamic IP, or DIPP.



Configure destination NAT

Translate a reachable address to an internal server, with optional port translation.



Write Security rules that support NAT

Match on pre-NAT addresses and post-NAT zones, then verify in the Traffic log.



LESSON 1

Network Address Translation

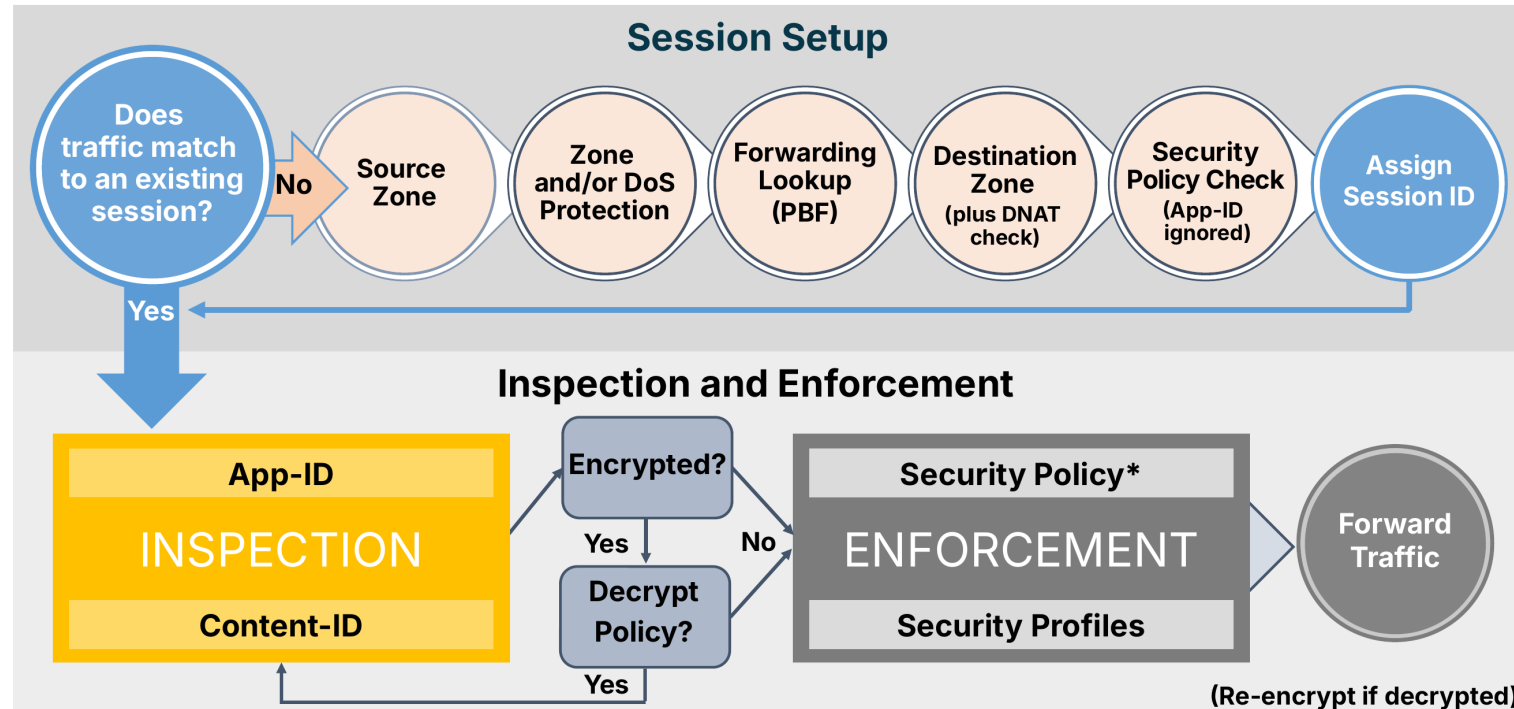
Warm-Up: Flow Logic of the Firewall

What are the four main steps the firewall goes through when it checks traffic?

Stage	What the firewall does
Session setup	Finds the zones, applies zone and DoS protection, looks up forwarding, assigns a session ID
Inspection	App-ID identifies the application; Content-ID examines the payload
Enforcement	Applies Security policy and Security Profiles
Forward	Re-encrypts if it decrypted, then sends the traffic on

Full detail: search the Palo Alto Networks Knowledge Base for "Packet Flow Sequence in PAN-OS".

Where NAT Fits in the Flow



*Policy check relies on pre-NAT IP addresses

Two touch points

Evaluated: during session setup, at the destination zone step.

Applied: at the end, as the traffic is forwarded.

In between, the packet still carries its **original** addresses.

Evaluated Early, Applied Late

- A **NAT policy rule** only translates addresses. It never allows or denies a packet.
- The firewall decides the translation early, so it already knows the **post-NAT zone** the packet will leave through.
- Nothing is rewritten until the end, so Security policy sees the **pre-NAT IP addresses**.
- A **Security policy rule** must still allow the traffic that the NAT rule translates.

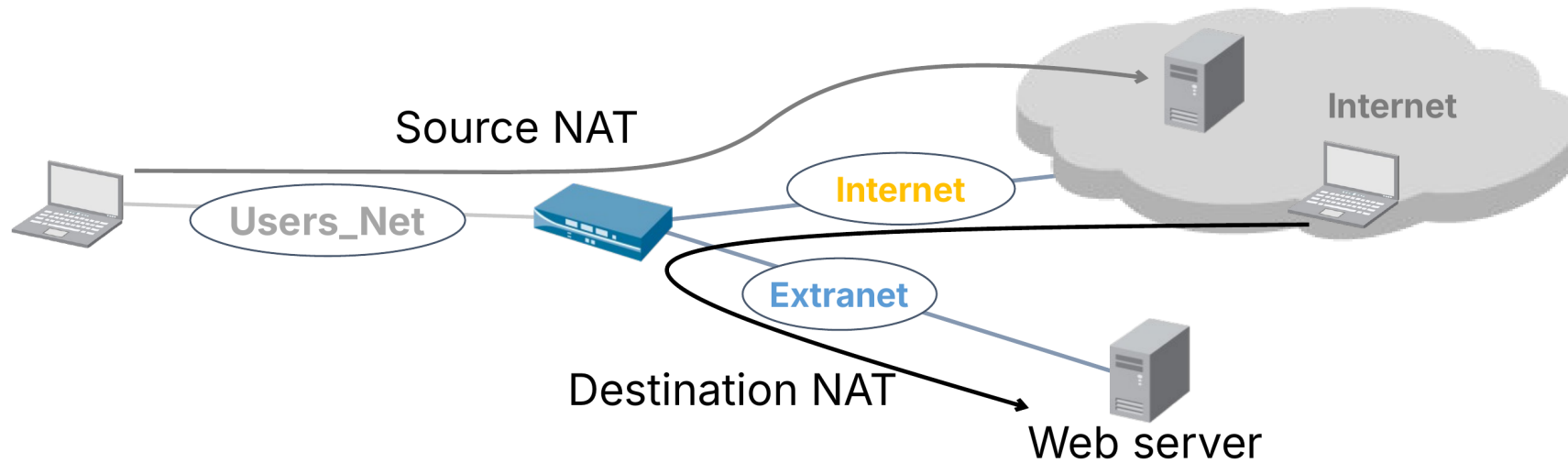


Say it until it sticks

Security policy matches **pre-NAT addresses** and **post-NAT zones**.

You will use this sentence in both halves of the lab.

NAT Types



Source NAT

- Private (internal) users reaching the public internet
- Usually outbound traffic

Destination NAT

- Public (external) hosts reaching a private internal server
- Usually inbound traffic

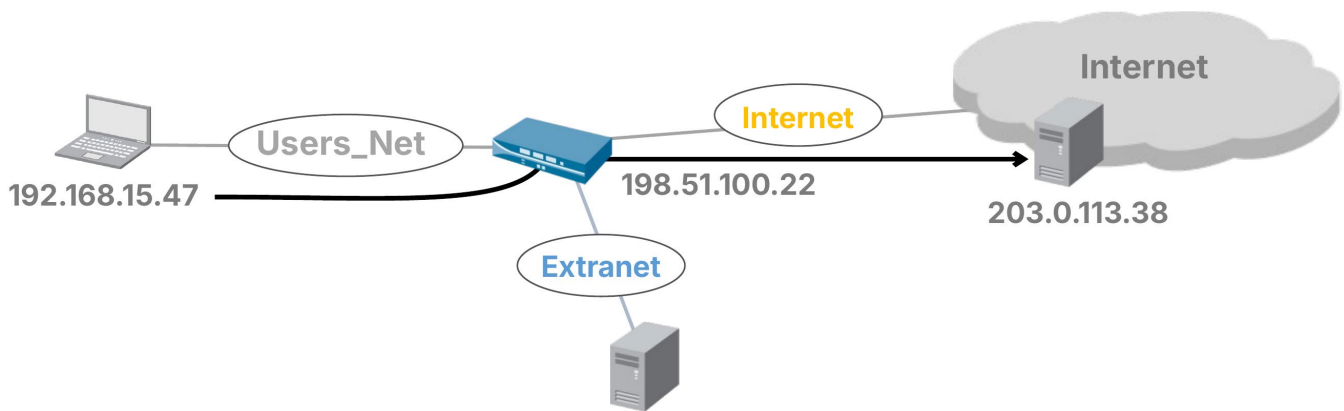
LESSON 2

Source NAT



How Source NAT Works

Source NAT translates an original source IP address to an alternate source IP address.



	Before	After
Source	192.168.15.47	198.51.100.22
Destination	203.0.113.38	203.0.113.38

- The firewall swaps in its own **Internet interface address** and remembers the mapping for the replies.
- Without it, the ISP would discard the traffic, because a **private address** is not routable on the internet.

Source NAT Types

Three ways to provision the translated address pool:

Type	Mapping	Source port	Typical use
Static IP	Fixed 1-to-1	Unchanged	A host that needs its own public address
Dynamic IP	1-to-1, next available address in a pool	Unchanged	A few public addresses handed out as needed
Dynamic IP and port (DIPP)	Many clients share one address	Translated	User access to the internet

Static IP is the only type that supports the Bi-directional option. A dynamic IP pool can be exhausted when more hosts than addresses are active.

Configure Source NAT

Original Packet tab = match criteria

NAT Policy Rule ?

General | **Original Packet** | Translated Packet

☐ Any	Destination Zone	☐ Any	☒ Any
☒ SOURCE ZONE ^	Internet	☒ SOURCE ADDRESS ^	☒ DESTINATION ADDRESS ^
☒ Users_Net		☒ 192.168.15.0/24	
	Destination Interface		
	any		
	Service		
	any		
+ Add - Delete		+ Add - Delete	+ Add - Delete

Translated Packet tab = translation

NAT Policy Rule ?

General | Original Packet | **Translated Packet**

Source Address Translation

Translation Type

Dynamic IP And Port

Address Type

Interface Address

Interface

ethernet1/1

IP Address

198.51.100.22/24

Destination Address Translation

Translation Type

None

- **Policies > NAT:** rules are checked top-down and the first match wins.
- For a source-only rule, leave **Destination Address Translation** set to None.

Examples: Static IP and Dynamic IP

Static 1:1 translation

	NAME	TAGS	Original Packet						Translated Packet	
			SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION	DESTINATION TRANSLATION
1	Source_Users_Net_to_Internet	Internet	 Users_Net	 Internet	any	 192.168.15.47	any	any	static-ip 198.51.100.22 bi-directional: yes	none

Dynamic IP translation

	NAME	TAGS	Original Packet						Translated Packet	
			SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINA... ADDRESS	SERVICE	SOURCE TRANSLATION	DESTINATION TRANSLATION
1	Source_Users_Net_to_Internet	Internet	 Users_Net	 Internet	any	 192.168.15.2-192.168.15.50	any	any	dynamic-ip 198.51.100.102-198.51.100.150	none

Pool exhaustion: when every dynamic IP address is in use, new sessions are blocked unless the rule is set to fall back to DIPP.

Dynamic IP and Port (DIPP)

One address, many clients: each session gets a different source port.

	NAME	TAGS	Original Packet						Translated Packet	
			SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINA... ADDRESS	SERVICE	SOURCE TRANSLATION	DESTINATION TRANSLATION
1	Source_Users_Net_to_Internet	Internet	 Users_Net	 Internet	any	 192.168.15.0/24	any	any	dynamic-ip-and-port 198.51.100.22-198.51.100.23	none

- Used when there are more internal hosts than **public IP addresses**.
- The translated address can be one address, a range, a subnet, or an interface.
- Roughly **64,000 sessions** per pool address before oversubscription.

Address Type

Interface Address: use an address already on an external interface.

Translated Address: use a new address that is not on an interface.

DIPP NAT Oversubscription

The same translated IP and port can be reused when the destinations are different.

Session Settings ⓘ

☒ Rematch all sessions on config policy change

ICMPv6 Token Bucket Size

ICMPv6 Error Packet Rate (per sec)

☒ Enable IPv6 Firewalling

☐ Enable Jumbo Frame

☐ Enable DHCP Broadcast Session

NAT64 IPv6 Minimum Network MTU

NAT Oversubscription Rate

Platform Default ▼

ICMP Unreachable Packet Rate (per sec)

Platform Default

☒ Accelerated Aging

Accelerated Aging Threshold

1x

Accelerated Aging Threshold

2x

Accelerated Aging Scaling Factor

4x

Accelerated Aging Scaling Factor

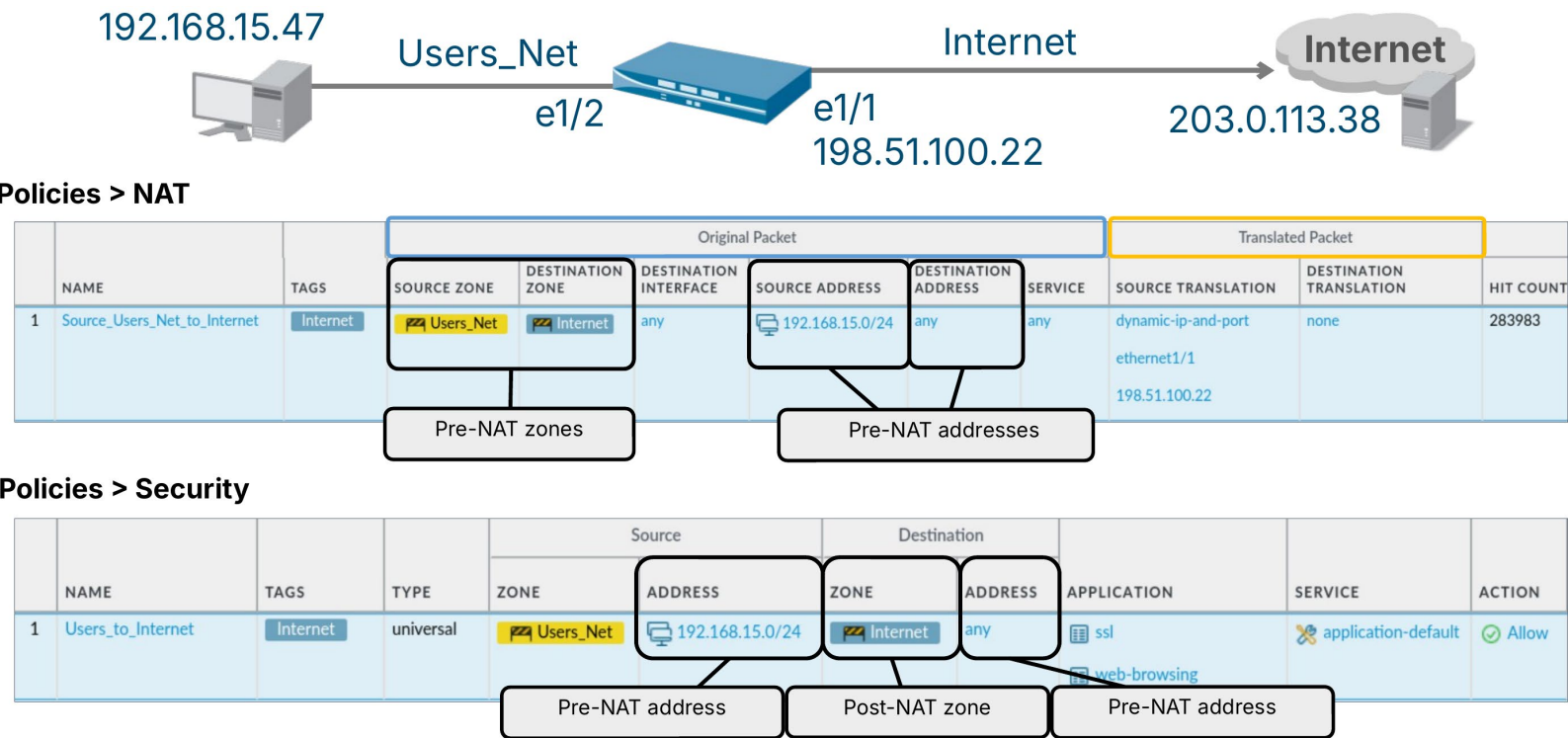
8x

☒ Packet Buffer Protection

Internal port	Firewall source	Destination
26435	198.51.100.22:25661	51.6.33.12
35435	198.51.100.22:25661	161.8.55.4
21569	198.51.100.22:25661	201.55.45.1
51043	198.51.100.22:25661	17.39.25.6

Capacity: concurrent sessions = oversubscription rate (8, 4, or 2) × address pool size. Set under Device > Setup > Session.

Source NAT and Security Policy



Reading the two rules

NAT rule: pre-NAT zones and pre-NAT addresses.

Security rule: pre-NAT addresses, post-NAT zone, action Allow.

Never write the Security rule against the translated address.

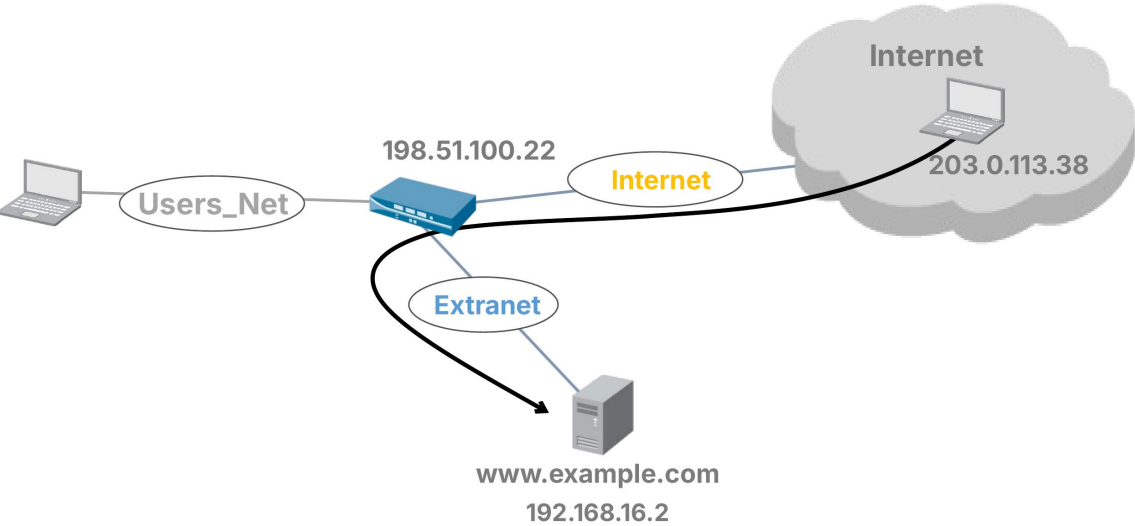
LESSON 3

Destination NAT



How Destination NAT Works

Destination NAT translates an original destination IP address to an alternate destination IP address.



	Before	After
Source	203.0.113.38	203.0.113.38
Destination	198.51.100.22	192.168.16.2
Dest. zone	Internet	Extranet

- Commonly used to make a server on a **private network** reachable from the public internet.
- After translating, the firewall does a **route lookup** to find the real egress interface and zone.

Configure Destination NAT

Original Packet tab = match criteria

NAT Policy Rule ?

General | **Original Packet** | Translated Packet

☐ Any	Destination Zone	☒ Any	☐ Any
☒ SOURCE ZONE ^	Internet	☒ SOURCE ADDRESS ^	☒ DESTINATION ADDRESS ^
☐ Internet			198.51.100.22
	Destination Interface		
	any		
	Service		
	service-http		
+ Add - Delete		+ Add - Delete	+ Add - Delete

Translated Packet tab = translation

NAT Policy Rule ?

General | Original Packet | **Translated Packet**

Source Address Translation

Translation Type None

Destination Address Translation

Translation Type Static IP

Translated Address 192.168.16.2

Translated Port [1 - 65535]

☐ Enable DNS Rewrite

Direction reverse

- The destination zone is the zone of the **pre-NAT address**, so it often equals the source zone.
- For a destination-only rule, leave **Source Address Translation** set to None.

Destination NAT Options

Static IP is the default; the other two solve specific problems.

Option	What it does	Use it when
Static IP	Fixed 1-to-1 translation; the destination port is unchanged	One reachable address maps to one server
Port translation	Also rewrites the destination port (Translated Port, 1 to 65535)	The server listens on a non-standard port, or one address fronts several servers
Dynamic IP	Translates to an FQDN, address object, or address group	The server gets its address from DHCP

If the Translated Port field is blank, the destination port is not changed.

Destination NAT Port Translation

Used when the destination server is listening on a port other than the well-known port.

NAT Policy Rule ?

General | Original Packet | **Translated Packet**

Source Address Translation

Translation Type None

Destination Address Translation

Translation Type Static IP

Translated Address 192.168.16.2

Translated Port 8000

☐ Enable DNS Rewrite

Direction reverse

service-http

The predefined `service-http` object includes TCP port 80 and TCP port 8080.

	NAME	TAGS	Original Packet						Translated Packet	
			SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION	DESTINATION TRANSLATION
1	Dest_NAT_From_Internet	Internet	Internet	Internet	any	any	198.51.100.22	service-http	none	destination-translation address: 192.168.16.2 port: 8000

Dynamic IP for Destination NAT

Translate to a destination host whose IP address is assigned by DHCP.

NAT Policy Rule

General

Original Packet

Translated Packet

Source Address Translation

Translation TypeNone

Destination Address Translation

Translation TypeDynamic IP (with session distribution)

Translated AddressPANW-Web-Server

Translated Port[1 - 65535]

Session Distribution MethodRound Robin

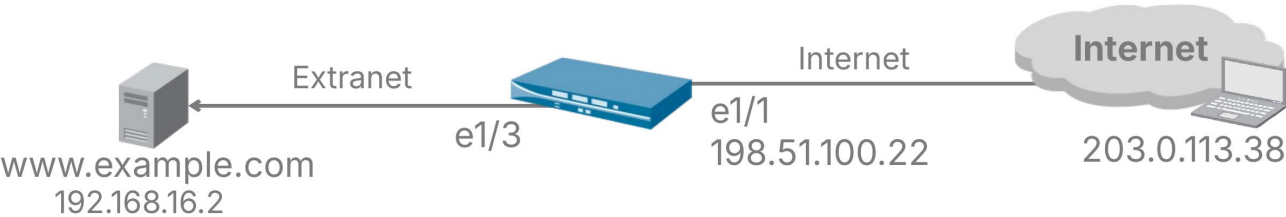
- The translated address can be an **FQDN**, an **address object**, or an **address group**.
- When the address changes, the NAT rule does not need to be edited.

More than one address?

If the FQDN resolves to several addresses, the firewall spreads sessions across them **round-robin**.

Up to 32 IPv4 and 32 IPv6 addresses per FQDN.

Destination NAT and Security Policy



Policies > NAT

Original Packet							Translated Packet	
NAME	TAGS	SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SOURCE TRANSLATION	DESTINATION TRANSLATION
1 Dest_NAT_From_Internet	Internet	Internet	Internet	any	any	198.51.100.22	none	destination-translation address: 192.168.16.2

Pre-NAT zones

Pre-NAT address

Policies > Security

NAME	TAGS	TYPE	Source		Destination		APPLICATION	SERVICE	ACTION
			ZONE	ADDRESS	ZONE	ADDRESS			
1 Web_Server_Access	Internet	universal	Internet	any	Extranet	198.51.100.22	web-browsing	application-default	Allow

Pre-NAT addresses

Post-NAT zone

Pre-NAT addresses

Where students slip

NAT rule zones: Internet to Internet.

Security rule zone: Internet to **Extranet**.

Security rule address: 198.51.100.22, not the server's private address.

Bi-directional Static NAT

One static source NAT rule that also translates the return direction.

NAT Policy Rule

General

Original Packet

Translated Packet

Source Address Translation

Translation TypeStatic IP

Translated Address198.51.100.22

☒ Bi-directional

Destination Address Translation

Translation TypeNone

- Lets an internal server **send and receive** traffic through the firewall.
- Creates an implicit **destination NAT** rule in the reverse direction.
- Available only for **static IP** translation.

Security Thread: NAT Is Not a Security Policy



SECURITY FOCUS

A NAT rule decides **where** a packet goes. Only the Security policy rule decides **whether** it goes, and with which application.

When you publish a server with destination NAT, write the Security rule as narrowly as the service allows: the pre-NAT address, the post-NAT zone, and the one application the server offers.

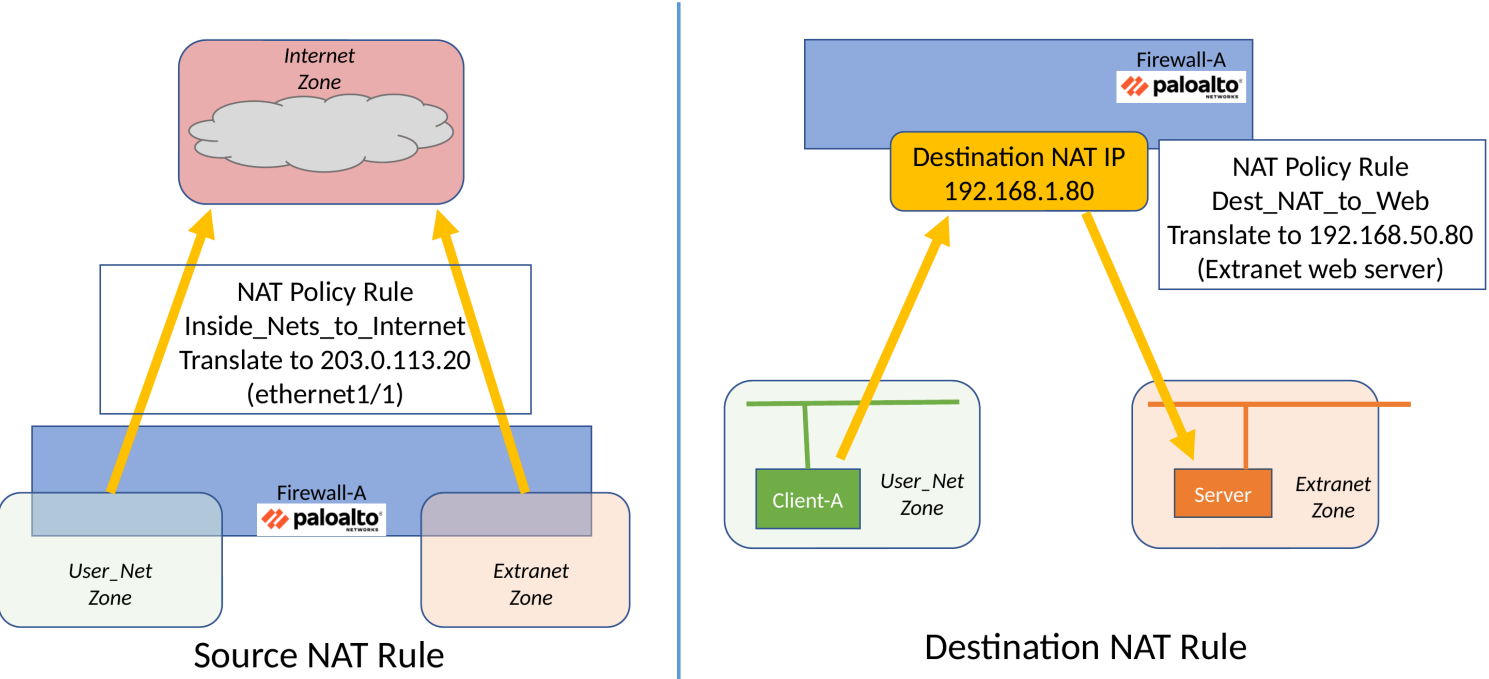
Why it matters: A destination NAT paired with a broad allow rule puts an internal server on the outside network with nothing limiting what reaches it.

Check Your Understanding

- NAT oversubscription is used with which translation type: static IP, dynamic IP, or dynamic IP and port?
- Which source NAT type changes the source IP address while leaving the source port unchanged?
- In the Security rule for a destination NAT, is the destination address the web server or the firewall interface address?
- True or false: source NAT is commonly used for private users to access the public internet.



Lab 6: What You Will Build



Two rules

Source NAT: Users_Net and Extranet out to the internet as 203.0.113.20.

Destination NAT: 192.168.1.80 to the web server at 192.168.50.80.

The lab guide names the destination rule Dest_NAT_To_Webserver.

Hands-On Lab: Creating and Managing NAT Policy Rules

NETLAB+ · PAN v11 Firewall Essentials, Lab 6 · baseline edu-210-11.0a-06.xml

- Load and commit the baseline configuration.
- Create `Inside_Nets_to_Internet` as a DIPP rule using the ethernet1/1 interface address.
- Commit, then verify with `ping 8.8.8.8` and a browser.
- Create `Dest_NAT_To_Webserver` as a static IP rule for 192.168.1.80.
- Commit, browse to `http://192.168.1.80`, and confirm both rules in the Traffic log.



Sign-in details are in the Lab Settings table of the lab guide.

Unit 7 Homework

- Always check **Blackboard** for assignments and when they are due.
- **Unit 7 Lab:** NETLAB+ Lab 6, Creating and Managing NAT Policy Rules
- **Unit 7 Quiz**



Key Takeaways



NAT policy is evaluated during session setup and applied as traffic is forwarded.



Source NAT gets private hosts out; DIPP shares one address across many clients.



Destination NAT publishes an internal server, with optional port translation.



Security policy matches pre-NAT addresses and post-NAT zones.



NAT translates; only a Security policy rule allows. Verify in the Traffic log.

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Questions?

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